

CENTRAL MINE PLANNING & DESIGN INSTITUTE (CMPDI)

Geological Drilling, Exploration & Project Planning Report (FY 2023-24)

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1. Exploration Drilling Performance & Stratigraphic Profiling

CMPDI deployed 154 mechanical and hydrostatic drilling rigs across Gondwana coal basins during the year. A total drilling meterage of 13.82 Lakh Metres was executed across 118 coal blocks in Jharkhand, Odisha, Chhattisgarh, Madhya Pradesh, West Bengal, and Maharashtra. Drilling operations encompassed 7.42 Lakh Metres in CIL leasehold blocks and 6.40 Lakh Metres in non-CIL/commercial auction coal blocks. Stratigraphic boreholes confirmed substantial coal reserves within the Lower Gondwana Barakar and Raniganj Formations.

Table 2: CMPDI Regional Institute Drilling Achievements (FY 2023-24)

Regional Institute	Headquarters	Target (Lakh m)	Achieved (Lakh m)	Geological Formations
RI-I	Asansol	1.25	1.31	Raniganj / Barakar
RI-II	Dhanbad	1.40	1.44	Barakar / Karharbari
RI-III	Ranchi	2.80	2.95	North & South Karanpura
RI-IV	Nagpur	1.50	1.52	Wardha Valley / Kamptee
RI-V	Bilaspur	3.20	3.38	Mand-Raigarh / Korba
RI-VI	Singrauli	1.80	1.86	Singrauli / Moher Basin
RI-VII	Bhubaneswar	1.30	1.36	Talcher / Ib Valley
CMPDI Total	Ranchi HQ	13.25	13.82	Gondwana Coal Basins

2. Coal Bed Methane (CBM) & Non-Conventional Gas Resources

CMPDI conducted detailed reservoir characterization and gas-in-place (GIP) estimations in Jharia and Bokaro coalfields. Estimated gas content within the deep seams (350m to 800m depth) ranged from 7.8 to 14.5 m³/tonne. Commercial CBM development plans for CIL leasehold blocks in Jharia (estimated GIP: 25.4 Billion Cubic Metres) were formalized in collaboration with national oil & gas exploration consortia, mitigating fugitive methane greenhouse emissions while supplying clean hydrocarbon fuels.

3. Detailed Project Reports (DPR) & Mine Feasibility Studies

During 2023-24, CMPDI prepared 28 Detailed Project Reports (DPRs) and Project Feasibility Studies with cumulative rated production capacity of 142.5 MTY. Highlights include the expansion DPR for Gevra OC (expanded capacity of 70 MTY, Asia's largest opencast coal project) and Kusmunda OC (50 MTY). Environmental impact assessments (EIA) and EMP formulations integrated numerical hydrogeological modeling, 3D laser scanner topography, and automated real-time dust monitoring sensor networks.